

MLX91804 Application Example with RF reporting compatible with VDA standard

1. Abstract

This document describes a firmware application example implemented by Melexis for the MLX91804/5 devices. The firmware provides RF reports compatible with VDA standard.

2. RFTx reports

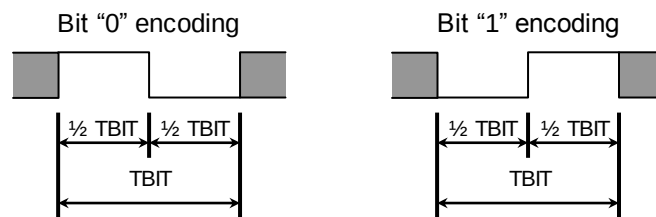
The firmware example implements periodical measurements of battery voltage, temperature and pressure with the next RF reporting (periodicity is about 4s).

Between the states when the MCU is active, the part stays in the deep sleep mode to save energy. If temperature change exceeds 10°C, the RFTx calibration is done before reporting.

2.1. RF carrier frequency and data encoding

The table below specifies the low level parameters of the RF communication channel:

Carrier frequency	433,92 MHz
Modulation type	FSK programmed approximately with ± 35 kHz deviation
Data encoding type	Manchester
Raw data rate	19174 bits per second that corresponds to the standard bit rate 19200 bits per second



Manchester bits encoding used in the example with RFTX_BIT_INVERT=0

2.2. Reporting messages description

Each report includes a single RF telegram (frame) having a format compatible with VDA standard for zero Packet ID.

There is a structure of the report telegram (MSB is going first):

1 byte	1 byte	1 byte	4 bytes	1 byte	1 byte	1 byte	4 NRZ bits
Preamble	Sync_word	Packet ID	Module ID	Pressure	Temperature	CRC8	Termination

Where:

Preamble is a 1 byte field with value 0xFF, Manchester coded.

Sync_word is a 1 byte field with value 0xF2, Manchester coded.

Packet ID field includes 1 byte identifying the payload format. In this example the Packet ID is zero that specifies a simplest pressure and temperature report.

Module ID field includes 4 bytes of the unique chip ID of the MLX91804 part mixed with a Supplier ID field in the highest nibble.

Pressure is an unsigned 8 bits value (1 byte) with pressure data.

"Pressure" byte can be converted to the [kPa] scale by using formulas below:

If "Pressure" = 0x00: measurement error happened
 If "Pressure" = 0x01: Pressure_kPa < 100 kPa (underflow)
 If "Pressure" = 0x02: Pressure_kPa = 100 kPa
 If "Pressure" = 0xFE: Pressure_kPa = 730 kPa
 If "Pressure" = 0xFF: Pressure_kPa > 730 kPa (overflow)
 Else Pressure_kPa = 100 + 2.5 * ("Pressure" - 2), expressed in [kPa]

Temperature is an unsigned 8 bits value (1 byte) with temperature data.

"Temperature" byte can be converted to the [°C] scale by using formulas below:

If "Temperature" = 0x00: measurement error happened
 If "Temperature" = 0x01: Temperature_degC < -40°C
 If "Temperature" = 0xF8: Temperature_degC > 125°C
 Else Temperature_degC = "Temperature" - 52, expressed in [°C]

2.3. CRC calculation

CRC8

"CRC8" is 8-bit CRC result generated by MLX91804 HW according to the Baicheva's 8-bit polynomial = $x^8 + x^5 + x^3 + x^2 + x + 1$ (0x12F) with preload value 0xAA.

A C-function below can be helpful at the receiver side to calculate such kind of CRC:

```
/* -----
 * 8-bit CRC calculation
 * Arguments:  uint8 const *data - pointer to the buffer with data covered by CRC
 *              uint8 len      - number of bytes to be covered by CRC
 * Returns:    The computed 8-bit CRC value of the buffer's data.*/

static const unsigned char CRC8_Table[256] = {
/*0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F */
0x00, 0x2f, 0x5e, 0x71, 0xbc, 0x93, 0xe2, 0xcd, 0x57, 0x78, 0x09, 0x26, 0xeb, 0xc4, 0xb5, 0x9a,
0xae, 0x81, 0xf0, 0xdf, 0x12, 0x3d, 0x4c, 0x63, 0xf9, 0xd6, 0xa7, 0x88, 0x45, 0x6a, 0x1b, 0x34,
0x73, 0x5c, 0x2d, 0x02, 0xcf, 0xe0, 0x91, 0xbe, 0x24, 0x0b, 0x7a, 0x55, 0x98, 0xb7, 0xc6, 0xe9,
0xdd, 0xf2, 0x83, 0xac, 0x61, 0x4e, 0x3f, 0x10, 0x8a, 0xa5, 0xd4, 0xfb, 0x36, 0x19, 0x68, 0x47,
0xe6, 0xc9, 0xb8, 0x97, 0x5a, 0x75, 0x04, 0x2b, 0xb1, 0x9e, 0xef, 0xc0, 0x0d, 0x22, 0x53, 0x7c,
0x48, 0x67, 0x16, 0x39, 0xf4, 0xdb, 0xaa, 0x85, 0x1f, 0x30, 0x41, 0x6e, 0xa3, 0x8c, 0xfd, 0xd2,
0x95, 0xba, 0xcb, 0xe4, 0x29, 0x06, 0x77, 0x58, 0xc2, 0xed, 0x9c, 0xb3, 0x7e, 0x51, 0x20, 0x0f,
0x3b, 0x14, 0x65, 0x4a, 0x87, 0xa8, 0xd9, 0xf6, 0x6c, 0x43, 0x32, 0x1d, 0xd0, 0xff, 0x8e, 0xa1,
0xe3, 0xcc, 0xbd, 0x92, 0x5f, 0x70, 0x01, 0x2e, 0xb4, 0x9b, 0xea, 0xc5, 0x08, 0x27, 0x56, 0x79,
0x4d, 0x62, 0x13, 0x3c, 0xf1, 0xde, 0xaf, 0x80, 0x1a, 0x35, 0x44, 0x6b, 0xa6, 0x89, 0xf8, 0xd7,
0x90, 0xbf, 0xce, 0xe1, 0x2c, 0x03, 0x72, 0x5d, 0xc7, 0xe8, 0x99, 0xb6, 0x7b, 0x54, 0x25, 0x0a,
0x3e, 0x11, 0x60, 0x4f, 0x82, 0xad, 0xdc, 0xf3, 0x69, 0x46, 0x37, 0x18, 0xd5, 0xfa, 0x8b, 0xa4,
0x05, 0x2a, 0x5b, 0x74, 0xb9, 0x96, 0xe7, 0xc8, 0x52, 0x7d, 0x0c, 0x23, 0xee, 0xc1, 0xb0, 0x9f,
0xab, 0x84, 0xf5, 0xda, 0x17, 0x38, 0x49, 0x66, 0xfc, 0xd3, 0xa2, 0x8d, 0x40, 0x6f, 0x1e, 0x31,
0x76, 0x59, 0x28, 0x07, 0xca, 0xe5, 0x94, 0xbb, 0x21, 0x0e, 0x7f, 0x50, 0x9d, 0xb2, 0xc3, 0xec,
0xd8, 0xf7, 0x86, 0xa9, 0x64, 0x4b, 0x3a, 0x15, 0x8f, 0xa0, 0xd1, 0xfe, 0x33, 0x1c, 0x6d, 0x42};

unsigned char CalcCRC8(unsigned char const *data, unsigned char len)
{
    unsigned char crc = 0xAA;

    while ((len--) != 0)
    {
        crc ^= *data;
        crc = CRC8_Table[crc];
        data++;
    }
    return crc;
} /* End of CalcCRC8 */
/* -----*/
```

Termination

field includes 4 NRZ coded bits that violate Manchester stream and hence can be used as a Stop sign for RF receiver.

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